

HW 2: Types of Data

CPE232 Data Models

Import Dependency

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
```

Part 1 : Basic Python

Task 1

Perform the following:

- Write a program to get input of different types from the user.
- Display type of each variable.
- Convert a variable of the type float to Integer.
- Demonstrate a comparison of before vs after the type conversion.

```
In [2]: # Input values from the user
string_value = input("Enter a string: ")
integer_value = int(input("Enter an integer: "))
float_value = float(input("Enter a float: "))
```

```
In [3]: # Display the data types
print(f"{string_value} {type(string_value)}")
print(f"{integer_value} {type(integer_value)}")
print(f"{float_value} {type(float_value)}")
```

```
Chanon <class 'str'>
13 <class 'int'>
12.23 <class 'float'>
```

```
In [4]: # Convert integer to float and vice versa
int_float_ver = float(integer_value)
float_integer_ver = int(float_value)
```

```
In [5]: # Show before and after convert float to int
print("Convert integer to float")
print(f"Before Convert: {integer_value} {type(integer_value)}")
print(f"After Convert: {int_float_ver} {type(int_float_ver)}")

print("Convert float to integer")
print(f"Before Convert: {float_value} {type(float_value)}")
print(f"After Convert: {float_integer_ver} {type(float_integer_ver)}")
```

```

Convert integer to float
Before Convert: 13 <class 'int'>
After Convert: 13.0 <class 'float'>
Convert float to integer
Before Convert: 12.23 <class 'float'>
After Convert: 12 <class 'int'>

```

Task 2

Given a dictionary of students with their scores, find the average score of each student across all subjects and identify the student with the highest average score.

```

In [6]: students_grades = {
        "John": [85, 90, 78],
        "Alice": [88, 92, 80],
        "Bob": [75, 85, 72],
        "Diana": [90, 95, 94],
        "Charlie": [70, 65, 80]
        }

```

```

In [7]: # Calculate and print the average grade for each student
avg_grades = {}
for person in students_grades:
    avg = sum(students_grades[person]) / len(students_grades[person])
    print(f"{person} : {avg:.2f}")
    avg_grades[person] = avg

```

```

John : 84.33
Alice : 86.67
Bob : 77.33
Diana : 93.00
Charlie : 71.67

```

```

In [8]: # Find the student with the highest average grade
highavg = None
for person in avg_grades :
    if highavg is None or avg_grades[highavg] < avg_grades[person] :
        highavg = person
print(f"{highavg} is the highest average grade and the average grade is {avg_grades[highavg]}")

```

Diana is the highest average grade and the average grade is 93.00

Part 2 : Working with CSV!

Add a new column to this CSV file named "Bonus," and calculate the bonus for employees in the Sales department as 10% of their MonthlyRate.

```

In [9]: # Load the CSV file
file_path = 'employee_data.csv'
df = pd.read_csv(file_path)

```

```

In [10]: df.head()

```

Out[10]:

	Age	DailyRate	Department	EducationField	Gender	MaritalStatus	MonthlyRate
0	41	1102	Sales	Life Sciences	Female	Single	19479
1	49	279	Research & Development	Life Sciences	Male	Married	24907
2	37	1373	Research & Development	Other	Male	Single	2396
3	33	1392	Research & Development	Life Sciences	Female	Married	23159
4	27	591	Research & Development	Medical	Male	Married	16632

In [11]: `# Add new column call 'Bonus'`
`df['Bonus'] = 0`
`df`

Out[11]:

	Age	DailyRate	Department	EducationField	Gender	MaritalStatus	MonthlyRate
0	41	1102	Sales	Life Sciences	Female	Single	1947
1	49	279	Research & Development	Life Sciences	Male	Married	2490
2	37	1373	Research & Development	Other	Male	Single	239
3	33	1392	Research & Development	Life Sciences	Female	Married	2315
4	27	591	Research & Development	Medical	Male	Married	1663
...	...	...	...	...	...	...	...
1465	36	884	Research & Development	Medical	Male	Married	1229
1466	39	613	Research & Development	Medical	Male	Married	2145
1467	27	155	Research & Development	Life Sciences	Male	Married	517
1468	49	1023	Sales	Medical	Male	Married	1324
1469	34	628	Research & Development	Medical	Male	Married	1022

1470 rows × 9 columns

In [12]: `# Calculate 10% of bonus into the 'Bonus' Column`
`df['Bonus'] = df['MonthlyRate'] * 1.1`

```
In [13]: # Print and show result
df.head()
```

```
Out[13]:
```

	Age	DailyRate	Department	EducationField	Gender	MaritalStatus	MonthlyRate
0	41	1102	Sales	Life Sciences	Female	Single	19479
1	49	279	Research & Development	Life Sciences	Male	Married	24907
2	37	1373	Research & Development	Other	Male	Single	2396
3	33	1392	Research & Development	Life Sciences	Female	Married	23159
4	27	591	Research & Development	Medical	Male	Married	16632

```
In [14]: # Save the updated DataFrame back to a CSV file
df.to_csv('employee_data_bonus.csv')
```

Part 3 : Working with Matplotlib

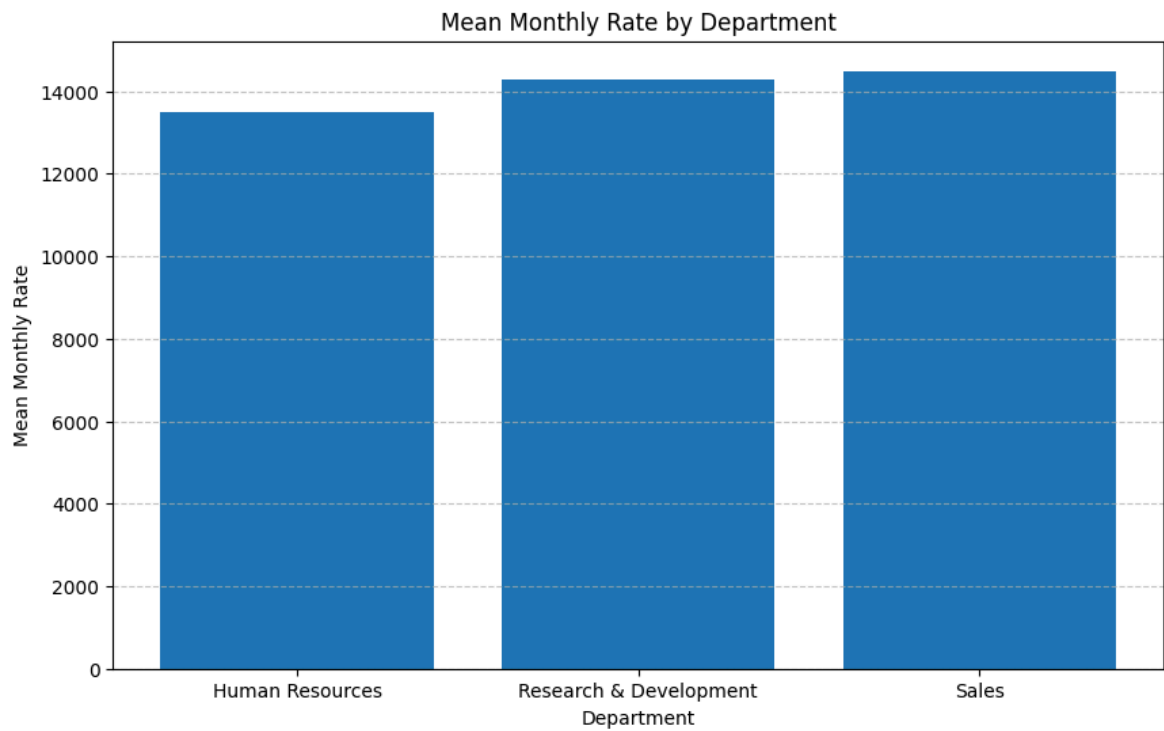
Plot a graph showing the average salary of employees in each department to compare the average salaries across departments.

```
In [15]: # Load CSV and create DataFrame
data = pd.read_csv('employee_data.csv')
df = pd.DataFrame(data)
```

```
In [16]: # Calculate average MonthlyRate by Department
avg_MonthlyRate = df.groupby("Department").agg({
    "MonthlyRate" : "mean"
})
print(avg_MonthlyRate)
```

	MonthlyRate
Department	
Human Resources	13492.984127
Research & Development	14284.865765
Sales	14489.793722

```
In [17]: plt.figure(figsize=(10, 6))
plt.bar(avg_MonthlyRate.index, avg_MonthlyRate['MonthlyRate'])
plt.xlabel('Department')
plt.ylabel('Mean Monthly Rate')
plt.title('Mean Monthly Rate by Department')
plt.grid(axis='y', linestyle='--', alpha=0.7)
```



Challenge!!! : Working with SATAN (Optional)

Great job, for not giving up on this subject!



Here's the story: The CSV file that was given to you isn't just an ordinary file -- it's data from an organization where Alya-San works!



Alya feels that something suspicious is going on within the company. She suspects that there's inequality in salaries between male and female employees.

She also wonders if the government might be intervening to encourage population growth by secretly increasing the salaries of married employees compared to those who are single.

Additionally, how much of a difference is there in salaries between employees who graduated from different fields of study?

Can you help Alya prove whether her suspicions are just in her head or if they're actually true?

For Aria, everyone is truly **e q u a l**.

```
In [35]: # Implement code here
avg_MonthlyRate_Gender = df.groupby(["Department", "Gender"]).agg({
    "MonthlyRate" : "mean"
})
print("Average Monthly Rate by Gender")
print(avg_MonthlyRate_Gender)
```

Average Monthly Rate by Gender		MonthlyRate
Department	Gender	
Human Resources	Female	14410.500000
	Male	13066.232558
Research & Development	Female	14932.350923
	Male	13863.221649
Sales	Female	14185.682540
	Male	14713.439689

```
In [36]: avg_MonthlyRate_Edu = df.groupby(["Department", "EducationField"]).agg({
    "MonthlyRate" : "mean"
})
print("Average Monthly Rate by Education Field")

print(avg_MonthlyRate_Edu)
```

Average Monthly Rate by Education Field

Department	EducationField	MonthlyRate
Human Resources	Human Resources	14810.740741
	Life Sciences	12813.875000
	Medical	12668.230769
	Other	9275.000000
	Technical Degree	13158.500000
Research & Development	Life Sciences	14594.704545
	Medical	14163.603306
	Other	13051.765625
	Technical Degree	14142.393617
Sales	Life Sciences	14523.786667
	Marketing	14076.943396
	Medical	15077.625000
	Other	15004.400000
	Technical Degree	14522.029412

```
In [37]: avg_MonthlyRate_MaritalStatus = df.groupby(["Department", "MaritalStatus"]).agg({
    "MonthlyRate" : "mean"
})
print("Average Monthly Rate by Marital Status")
print(avg_MonthlyRate_MaritalStatus)
```

Average Monthly Rate by Marital Status

Department	MaritalStatus	MonthlyRate
Human Resources	Divorced	14277.125000
	Married	13010.588235
	Single	13789.538462
Research & Development	Divorced	14337.553571
	Married	13981.224018
	Single	14678.532895
Sales	Divorced	14245.425287
	Married	14348.155340
	Single	14819.450980

Since Kobayashi-san will be reviewing your code, please present the data in a way that is clear and accurate. Otherwise, Kobayashi-san might fire you from the company!



```
In [44]: # Write your code here
chart_data = avg_MonthlyRate_Gender.unstack()
```

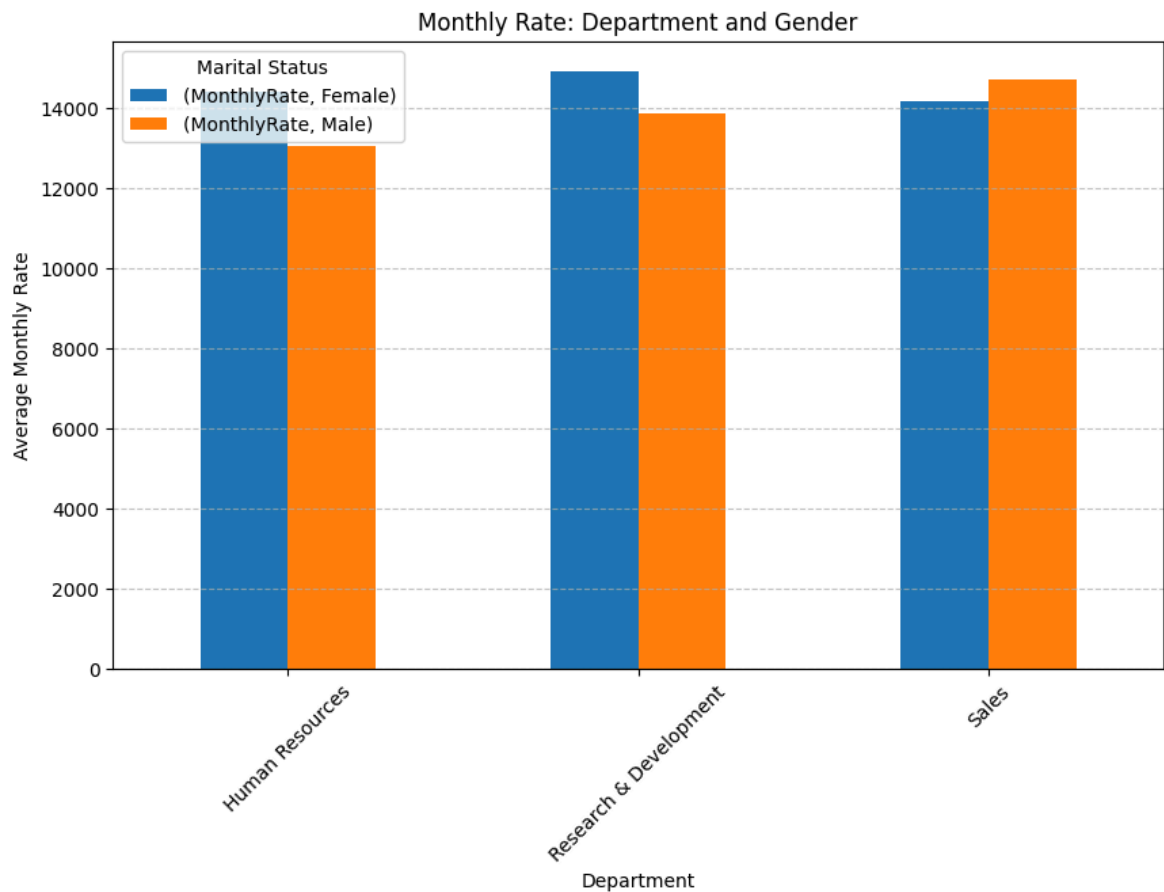
```

chart_data.plot(kind='bar', figsize=(10, 6))

plt.title('Monthly Rate: Department and Gender')
plt.ylabel('Average Monthly Rate')
plt.xlabel('Department')
plt.xticks(rotation=45)
plt.legend(title='Marital Status')
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()

```



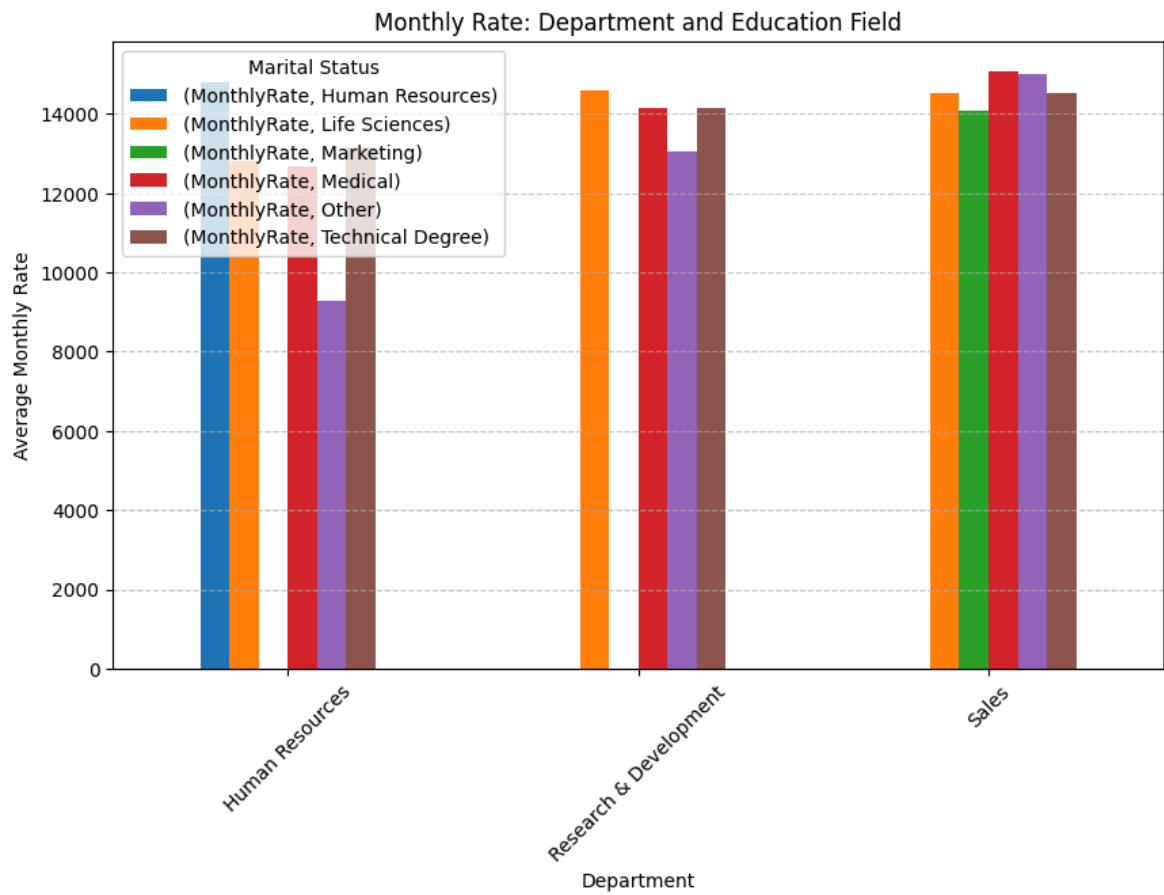
```

In [45]: chart_data = avg_MonthlyRate_Edu.unstack()
chart_data.plot(kind='bar', figsize=(10, 6))

plt.title('Monthly Rate: Department and Education Field')
plt.ylabel('Average Monthly Rate')
plt.xlabel('Department')
plt.xticks(rotation=45)
plt.legend(title='Marital Status')
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()

```

```
In [46]: chart_data = avg_MonthlyRate_MaritalStatus.unstack()
chart_data.plot(kind='bar', figsize=(10, 6))

plt.title('Monthly Rate: Department vs. Marital Status')
plt.ylabel('Average Monthly Rate')
plt.xlabel('Department')
plt.xticks(rotation=45)
plt.legend(title='Marital Status')
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

